Project Initialization and Planning Phase

Project Proposal (Proposed Solution)

Metadata Field	Value Details
Date	19 October 2026
Team ID	
Project Title	Health360: Clinical & Healthcare Risk Analysis
Maximum Marks	3 Marks

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview

Dimension	Specification & Context
Objective	Develop an interactive Business Intelligence and Clinical Risk Analytics solution (Health360) using Tableau Public and responsive web technologies to analyze 5,000 patient records across 6 age demographics. The primary objective is to evaluate blood pressure variations, chronic fatigue patterns, and lifestyle risk factors (exercise tiers, smoking, alcohol) to enable early diagnosis and data-driven preventive interventions.

Dimension	Specification & Context
Scope	Covers descriptive, diagnostic, and risk correlation analysis of 5,000 anonymized patient records across six demographic cohorts (Teenager, Young Adult, Adult, Middle Age, Old, Senior). The analytical scope encompasses systolic BP benchmarking (~134.4 mmHg mean), fatigue prevalence (50.8%), physical activity tiers (Levels 0, 1, 2), substance exposure, and a normalized health risk score matrix (mean 100.0). The deployment scope includes an interactive Tableau Public workbook and a responsive web dashboard deployed via GitHub Pages.

Problem Statement

Aspect	Analysis & Strategic Impact
Description	Healthcare practitioners and public health analysts face massive fragmentation when evaluating patient risk profiles across clinical spreadsheets. Vital physiological parameters (blood pressure, chronic fatigue) and critical lifestyle drivers (substance abuse, physical inactivity) are recorded in disconnected formats, making it extremely difficult to detect early cardiovascular risk spikes and demographic-specific health deterioration.
Impact	Solving this problem eliminates diagnostic blind spots, accelerates clinical decision-making by over 70%, and identifies high-risk cohorts before acute cardiovascular or chronic clinical episodes develop. It empowers public health

Aspect	Analysis & Strategic Impact
	authorities to deploy targeted community interventions, optimize clinical resource allocation, and lower long-term patient morbidity and hospitalization costs.

Proposed Solution

Strategic Focus	Detailed Plan & Deliverables
Approach	**End-to-end data intelligence methodology:**1. *Data Ingestion & Cleansing:* Ingestion and validation of 5,000 clinical records from `smart_healthcare_dataset.csv`.2. *Analytical Processing:* Calculating demographic-stratified KPIs, risk variances, and cross-variable correlations using Python and Tableau calculated fields.3. *Visual Analytics:* Designing 6 specialized clinical charts and heatmaps.4. *Cloud Deployment:* Building an executive 5-point clinical data story and deploying a responsive Chart.js web dashboard via GitHub Pages.
Key Features	1. **Multi-Cohort Blood Pressure Profiling:** Demographic systolic variance analysis.2. **Fatigue & Symptom Prevalence Matrix:** 50.8% demographic mapping.3. **Substance Consumption Dual-Bar Analytics:** Comparative smoking & alcohol correlation.4. **6×3 Multi-Tier Lifestyle Risk Heatmap:** Interactive risk visualization across cohorts.5. **Guided Clinical Data Story:** 5-point executive story for fast decision-making.6. **Responsive Web Portal:**

Strategic Focus	Detailed Plan & Deliverables
	Cross-device accessibility with instant metric filters.

Resource Requirements

Category 1: Hardware

Resource	Technical Metric	Provisioning Detail
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 / AMD Ryzen 5 or higher (Quad-Core, 2.5 GHz+), standard integrated/dedicated GPU
Memory	RAM specifications	8 GB / 16 GB DDR4 RAM
Storage	Disk space for data, models, and logs	512 GB / 1 TB SSD (minimum 50 GB free disk space for datasets, workbooks, and local staging)

Category 2: Software

Software Type	Specification	Usage Purpose
Frameworks	Python frameworks	Python 3.10+, Pandas, NumPy, Matplotlib/Seaborn (for data preparation)
Libraries	Additional libraries	Chart.js (v4.x), Tailwind CSS (CDN), FontAwesome (icons)
Development Environment	IDE, version control	VS Code / Jupyter Notebook, Git, GitHub & GitHub Pages, Tableau Desktop Public Edition

Category 3: Data

Source	Size & Format	Structure Description
Data	Source, size, format	Kaggle / Healthcare Repository (`smart_healthcare_dataset.csv`), 5,000 patient records (~500 KB to 1 MB), Structured CSV format with 6 age cohorts and clinical vital indicators